

FOD System for EV Wireless Charging using LTspice and MATLAB

Foreign Object Detection for Electric Vehicle Dynamic Wireless Power Transfer Systems

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1. Project Overview

Wireless Power Transfer (WPT) systems for electric vehicles are increasingly being deployed for convenient, cable-free charging. However, these systems are vulnerable to foreign metallic objects — such as coins, bolts, or aluminium sheets — placed between the transmitter and receiver coils during charging.

These objects absorb energy through eddy currents, generating heat and posing a serious safety and reliability risk. This project develops a Foreign Object Detection (FOD) system for an EV WPT system through:

- Full SS (Series-Series) topology WPT circuit simulation in LTspice
- Physics-based FOD modeling using magnetically coupled inductor loops
- Wavelet Transform signal processing in MATLAB
- Feature extraction and threshold-based detection

2. System Architecture

2.1 WPT System Block Diagram

The complete system follows this signal flow:

DC Source (50V) → H-Bridge Inverter (15 kHz) → Primary Resonant Tank (C3 + L1)
→ Magnetic Coupling (K) → Secondary Resonant Tank (L2 + C4) → Load (R2)

2.2 FOD Model

A foreign metallic object is modeled as a third magnetically coupled inductor loop:

- L3 represents the eddy current inductance of the metal object
- R_fod represents the resistive loss due to eddy currents (heat dissipation)
- L3 and R_fod form a shorted loop — no external connection, purely magnetic
- Weak coupling K2 between L1 and L3 simulates the magnetic interaction

This approach accurately represents the reflected impedance of a real metallic object in the magnetic field of the transmitter coil, without requiring any electrical connection.

3. Circuit Parameters

Parameter	Value	Notes
Supply Voltage (Vcc)	50 V	DC input source
Switching Frequency	15 kHz	H-bridge gate signals
Primary Inductance (L1)	100 μ H	Transmitter coil
Secondary Inductance (L2)	100 μ H	Receiver coil
Resonant Capacitors (C3, C4)	1.126 μ F	Tuned to 15 kHz resonance
Coupling Coefficient (K)	0.3 – 0.69	TX-RX coil coupling
Load Resistance (R2)	10 Ω	Secondary load
Gate Drive Voltage	30 V	NMOS gate signal amplitude
MOSFET Model	MYMOS (Level-1)	VTO=2V, KP=100, LAMBDA=0.01
Inductor Series Resistance	0.01 Ω	Winding resistance (Rser)

3.1 Resonant Frequency Verification

The resonant frequency is calculated as:

$$f_r = 1 / (2\pi \times \sqrt{L \times C}) = 1 / (2\pi \times \sqrt{(100\mu \times 1.126\mu)}) \approx 15 \text{ kHz} \quad \checkmark$$

4. FOD Object Parameters

Object Type	L3	R_fod	K2 (coupling to L1)
Small metal (bolt)	10 μ H	1 Ω	0.1
Large metal (sheet)	100 μ H	0.1 Ω	0.3
Copper object	50 μ H	0.01 Ω	0.2

Different objects produce different disturbance signatures in I(L1), enabling classification by object type in future work.

5. MATLAB Signal Processing Pipeline

5.1 Workflow

- Export I(L1) waveform from LTspice as .txt file
- Load data into MATLAB using readmatrix()
- Apply Wavelet Decomposition: wavedec(signal, 5, 'db4')
- Extract detail coefficients at each level: detcoef()
- Compute wavelet energy per level: $\sum(d.^2)$
- Compare Normal vs FOD features
- Apply threshold: if energy > threshold → FOD Detected

5.2 Wavelet Transform Details

Parameter	Value
Wavelet Family	Daubechies 4 (db4)
Decomposition Levels	5
Features Extracted	RMS, Peak, Energy at L1–L5
Total Features per Sample	7

6. Results

6.1 Waveform Analysis

- I(L1) shows clean sinusoidal resonance at 15 kHz under normal conditions ($\pm 13A$ peak)
- I(L2) confirms successful power transfer to secondary side
- I(L3) shows eddy current flowing in FOD object when present
- H-bridge output confirmed as symmetric $\pm 50V$ square wave

6.2 Feature Comparison

Feature	Normal	FOD Present	Change
RMS Current	3.21 A	3.38 A	+5.64%
Peak Current	13.86 A	16.51 A	+23.64%
Wavelet Energy Level 1	383.75	583.40	+52.03%
Wavelet Energy Level 2	5279.37	7209.68	+36.56%
Wavelet Energy Level 3	15811.17	21709.45	+37.30%
Wavelet Energy Level 4	19599.45	23249.01	+18.62%

Wavelet Energy Level 5	21282.36	23716.86	+11.44%
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6.3 Key Insight

FOD presence causes a consistent and significant increase in wavelet energy across all decomposition levels, with the strongest signature at Level 1 (+52%). This is because eddy current interaction introduces high-frequency disturbance components into the primary coil current, which are captured by the wavelet detail coefficients at higher frequency bands.

7. Tools Used

Tool	Purpose
LTspice XVII / 24	Circuit simulation and waveform export
MATLAB	Signal processing and wavelet analysis
MATLAB Wavelet Toolbox	wavedec(), detcoef() functions

8. Future Work

- Add full-bridge rectifier on secondary side for DC output
- Reduce coupling coefficient to realistic EV WPT range ($K = 0.1\text{--}0.3$)
- Simulate multiple FOD scenarios with varied object parameters
- Train SVM classifier for multi-class FOD detection
- Target classification accuracy > 95% (reference paper achieved 98%)
- Validate detection system against hardware prototype

9. References

- Soft Fault Diagnosis of WPT System Main Circuit using Wavelet Transform and SVM — IEEE
- SAE J2954 — Wireless Power Transfer for Light-Duty Plug-in/Electric Vehicles
- Wireless Power Transfer for Electric Vehicles — IEEE Transactions on Power Electronics
- LTspice XVII User Guide — Analog Devices